

UNIVERSIDAD CARLOS III DE MADRID



**MATERIALS SCIENCE AND
ENGINEERING AND
CHEMICAL ENGINEERING**

**LABORATORY OF AEROSPACE
MATERIALS I**

Aerospace Engineering

MANDATORY RULES OF USE AND WORK IN THE LAB

GENERAL SAFETY RULES

SAFETY IN LABORATORIES. GENERAL MEASURES

In general:

- Read carefully the outline for the practice. Follow the teacher's instructions at all times. If you have any doubt, ask the responsible for the laboratory session.
- You cannot perform experiments which aren't authorized by the teacher.
- Behavior in the laboratory must be correct; you shouldn't make jokes, nor play.

Research is an important work at the University. Do not touch the equipment and installations that do not belong to your laboratory session.

Health:

- If you have any incompatibility with some chemical product, you have a pacemaker or any pathology which it could be relevant to the normal course of a practice, you must inform the teacher before doing it.

Clothing:

- During works with machines or nearby don't wear large clothes with parts that hung (for example avoid ties, fringes, or frayed edges, etc) It is recommended you don't wear rings, wristwatches, necklaces, or other accessories that can be hooked.
- You shouldn't wear short sleeves, short skirts, shorts, or sandals.
- If you have long hair, you must tie it in order to avoid risks.

Habits:

- Do not eat, drink, chew gum, nor apply cosmetics in the lab.
- Do not suck pens that have been rolling on the table. You could accidentally ingest chemical substances.
- Wash always your hands after each experiment.

Order and cleanliness:

- Keep always tidy and clean your place of work. The work table must be free of coats, bags and books.

In case of accident:

- You must immediately notify any incident to the teacher. Even small, it will be considered as an accident.
- You must know the location and operation of showers and fire extinguishers as well as escape routes.
- If it's necessary an evacuation from the lab, this shall be done in an orderly manner by any of the two doors with anti-panic lock that exist in it. Exit from the building will be done by the escape routes.
- The area located at the exterior of the building (next to the flags) is set as mandatory meeting point.

SPECIFIC SECURITY NORMS

RISK IN THE USE OF MACHINES:

- In some instances, risk at the source cannot be eliminated being necessary to use collective protection means: guards and security devices.
 1. A guard is the component of a machine used as a physical barrier to protect.
 2. A protective device is one that prevents starting or holding a dangerous machine stage as it is detected or the possible human presence in the danger zone.
- Do not turn off existing security devices.
- Use the protective measures correctly.
- Do not use equipment or machinery without knowing how it works.
- Before performing any task in a machine carefully follow the instructions that are given to you. In case of doubt, ask the teacher.
- Unplug tools and equipment prior to its setting.
- In case of danger or an unforeseen event, immediately click the emergency stop (the red mushroom type push button)
- It isn't allowed the use by students of presses, bending machines, thread grinders, drills, millers, lathes, or any machine tool.

It is forbidden to repair, clear, or clean any equipment by students. Notify the anomaly in order that trained personnel perform the task.

USE OF FURNACES:

- Be extremely cautious during handling of furnaces; if you observe any anomaly, immediately notify it to the teacher. Use protective gloves and goggles at all times. Make use of clamps for using graduated cylinders and test tubs. Don't leave gloves, coats, or any material that may be combustible near ovens.

USE OF THE TENSILE TEST MACHINE:

- Be extremely cautious placing the sample between the jaws. Samples must be placed when the system is turn off. For the extraction of the two pieces from the jaws, turn move to star position and when the system is turn off you can remove it.

OPERATING LAB NORMS

- a) The attendance to **ALL** sessions is **MANDATORY** to pass Laboratory Practices.
- b) Students should be prompt to enter the laboratory.
- c) Students must read the outline of each practice **BEFORE** coming into the lab.
- d) At the end of each practice you must tidy up and clean all the material, leaving it as it was at the beginning; the teacher will revise it and will stamp the corresponding card of each student.
- e) It isn't possible to change the group without justifying.
- f) Mobile phones must be silent during the practical sessions. In exceptional cases teacher must be consulted previously.
- g) Students must attend to lab practices wearing adequate clothes (long sleeves clothes, closed shoes) and lab coat. When necessary students will be provided with other protection means.
- h) Security in the laboratory is everyone's responsibility, so never do anything that will place your, or anyone else's, safety at risk.
- i) In case of doubt, **ALWAYS** ask the assistant, as misinformation is the main cause of accidents.
- j) Do not smoke, eat, or drink in the lab.

SESSION I: CRYSTALLINE STRUCTURE

AIM

Construction of closed packed and non compact crystalline structures. X-ray diffraction pattern indexing. Determination of lattice parameters of crystalline pure solids and mixtures.

MATERIALS

- 31 steel spheres
- 51 magnetic washers
- Several diffraction patterns
- Graduated ruler

INTRODUCTION

Crystalline lattices

Metallic solids, semiconductor materials, many ceramic materials and even some polymers have a crystalline structure where atoms display both a short-range and long order, and the spatial arrangement of the atoms extends to all the material forming a lattice. Physical properties of these solids can be anisotropic. Conversely, amorphous solids are isotropic, they have no regular reticular structure, and its properties are equal in all directions.

Crystalline structures can be easily described by the so-called unit cell, which is the smallest unit that, by indefinite repetition in the three spatial directions, generates the crystal. If we know the exact arrangement of atoms within a unit cell, we will know the arrangement of the whole crystal. The unit cell is always a parallelepiped (Figure 1) and considering the possible combinations of distances (a , b , c) from the points of the network on its axes and the angles (α , β , γ) between the axes, seven crystal systems can be generated. Any crystalline solid can be described by one of the seven crystallographic systems.

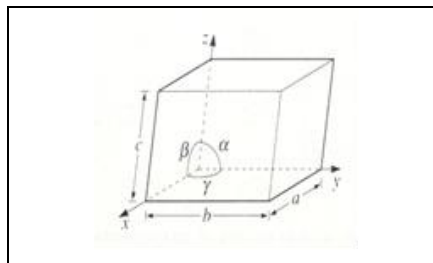


Figure 1.- Unit cell and cell parameters

The number of possibilities for the selection of the unit cell can be infinite, since any parallelogram whose sides connect network points is always a valid choice. The general approach is to choose the smallest one, called primitive cell, which only has points at the vertices of the elemental parallelepiped. It is described applying three unit translations a , b ,

c, and among the different options, the one that has most similar reticular parameters mutually forming angles of 90°, if any, is selected. But it can sometimes be preferable to choose a cell with more than one point associated to it, which is called a non-primitive or multiple. We can therefore describe various lattices, which are named with the symbols P, I, F, C (A, B), when they represent a primitive cell, a body centered cell, a cell centered on all its faces, and a cell centered on two opposite faces, respectively; in this last case, the symbols A, B, or C refer to cells centered on parallel faces YZ, XZ or XY, respectively. Bravais proved that there are only fourteen crystalline lattices associated with these seven crystal systems.

In a crystalline lattice we can draw, in the most diverse directions, series of infinite planes parallel and equidistant from each other, each containing linear sequences of reticular points. Every series of planes divides crystallographic axes in an integer of equal parts. The plane whose intersections with axes are a/h , b/k and c/l , where h , k , l , are integers without any common divisor, is the plane closest to the origin belonging to the same family of parallel and equidistant planes. Numbers h , k , l , identify the position and orientation of the plane relative to the crystallographic axes and are called Miller indices, written with the notation (hkl) . These are found directly reducing to the smallest integer the mutual values of fractional intersections with the crystallographic axes. When a plane is parallel to one of the axes, intercepts it at infinity and the corresponding Miller index is zero ($1/\infty = 0$). In the hexagonal system it is often employed a third index i , to designate completely the planes. The existence of this new index is imposed by the convenience of using a third axis, a_3 , coplanar with a_1 and a_2 . From this point of view, the planes in this system are designated with the indices $(hkil)$, calling them Miller-Bravais indices. However, the value of i is deducted directly from values h and k , since $h+k+i=0$.

The distance between two planes of parallel adjacent atoms with the same Miller indices hkl is known as inter-planar distance d and for a crystal structure of cubic symmetry is given by:

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2}$$

For systems with non cubic symmetry is given by:

$$\text{TETRAGONAL SYMMETRY} \quad \frac{1}{d^2} = \frac{h^2 + k^2}{a^2} + \frac{l^2}{c^2}$$

$$\text{ORTHORHOMBIC SYMMETRY} \quad \frac{1}{d^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$$

$$\text{HEXAGONAL SYMMETRY} \quad \frac{1}{d^2} = \left(\frac{h^2 + hk + k^2}{a^2} \right) + \frac{l^2}{c^2}$$

$$\text{MONOCLINIC SYMMETRY} \quad \frac{1}{d^2} = \frac{1}{\sin^2 \beta} \left(\frac{h^2}{a^2} + \frac{k^2 \sin^2 \beta}{b^2} + \frac{l^2}{c^2} - \frac{2hl \cos \beta}{ac} \right)$$

$$\text{TRICLINIC SYMMETRY} \quad \frac{1}{d^2} = \frac{1}{V^2} \left[h^2 b^2 c^2 \sin^2 \alpha + k^2 a^2 c^2 \sin^2 \beta + l^2 a^2 b^2 \sin^2 \gamma + 2hkabc^2 (\cos \alpha \cos \beta - \cos \gamma) + 2kla^2 bc (\cos \beta \cos \gamma - \cos \alpha) + 2hlab^2 c (\cos \alpha \cos \gamma - \cos \beta) \right]$$

Types of packing: building structures

The different crystalline systems describe how atoms, ions, or molecules are packed; when building the various types of structures it should be taken into account that the arrangement of particles will occur in a way such that the maximum density is achieved, considering both energetic and steric aspects. In particular, almost all metallic elements and many molecular crystals present these close packed structures. If we consider atoms or ions as rigid spheres, the best way to make use of space in two dimensions is that each sphere must be surrounded by other six, the closest neighbors. The infinite extension of this arrangement in two directions (xy) gives rise to compact packed sheets or layers (Fig. 2a). Another way of packing spheres less compactly is forming a square network (Fig. 2b).

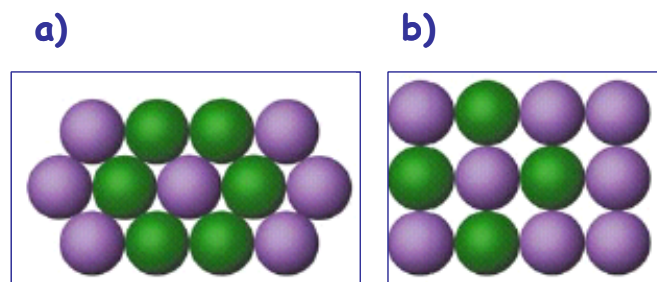


Figure 2.- Layer of close packing (a) and square packing (b).

The simplest manner to pile up compact layers is precisely putting the centre of the particle on the centre of the bottom layer. The result is a non compact packaging (Fig. 3a), called simple hexagonal structure. The way in which space is better used, and therefore the energetically more favorable, is to place the particles of the upper layer on the interstices of the bottom layer (Fig. 3b). This type of packing is called close packing.

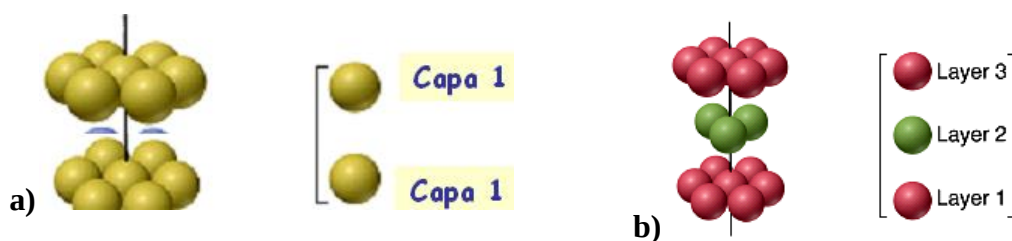


Figure 3.- Non compact hexagonal packing (a) and hexagonal close packing (b)

There are two options in placing the third layer:

i) To place it in the same interstitial holes as in the holes of the first layer, i.e., if A is the first layer and B is the second one, the third layer is placed as A, so the sequence of the

different layers would be: ABABA... The packing formed in this way is called hexagonal close packing (HCP).

ii) To place it in the interstitial holes of the second layer, without matching the arrangement of the first one. i.e., if we call C the third layer, the stacking sequence would be ABCABC.... The packing thus formed is called cubic close packing (CCP), Figure 5.

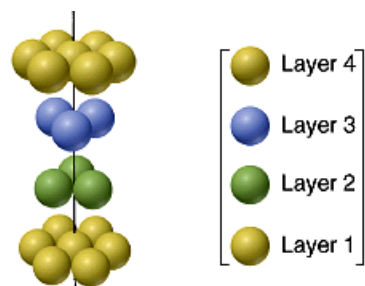


Figure 4.- Stacking sequence of layers in a cubic close packing structure.

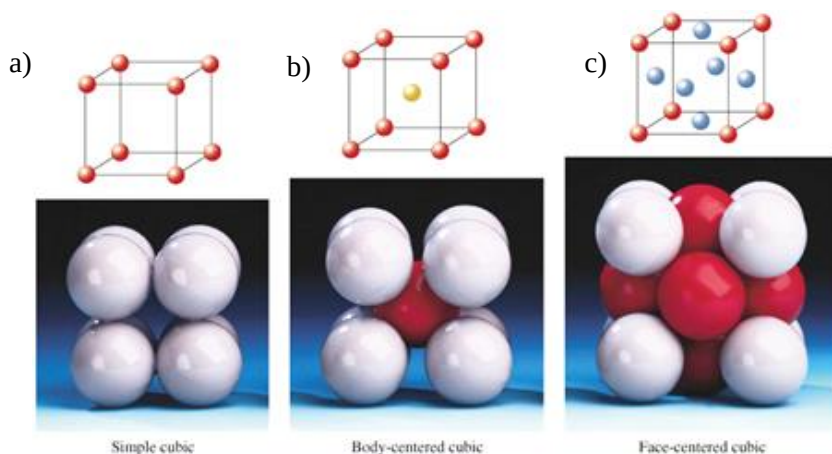


Figure 5.- Cubic structures.

Taking into account these two types of packing we can describe structures as:

Simple hexagonal: two close packing layers stacked in the sequence AA, (Fig. 3a).

Hexagonal close packing (HCP): two close packing layers stacked in the sequence ABAB, (Fig. 3b).

Simple cubic: two square packing layers stacked in the sequence AA, (Fig. 5a).

Face-centered cubic (FCC): three close packing layers stacked in the sequence ABCABC, (Fig. 5c).

A type of packing appears in many compounds and that cannot be constructed by stacking square or close packing layers. That is called the body-centered cubic (BCC) structure (Fig. 5b). This structure can be interpreted as a stacking in the sequence ABAB... of non compact square packing layers where the particles of each layer don't intercept each other.

Many structures of ionic crystals can be described alternatively as close packing of anions (more voluminous) with cations (smaller) occupying the holes or interstitial sites. If the anions are big enough to be in contact with each other, they will be arranged forming a

cubic close packing. In this arrangement, there are two types of interstitial sites, tetrahedral and octahedral sites. Thus, for example, NaCl structure can be described as a close packing of chloride ions in which all the octahedral sites are occupied by sodium ions.

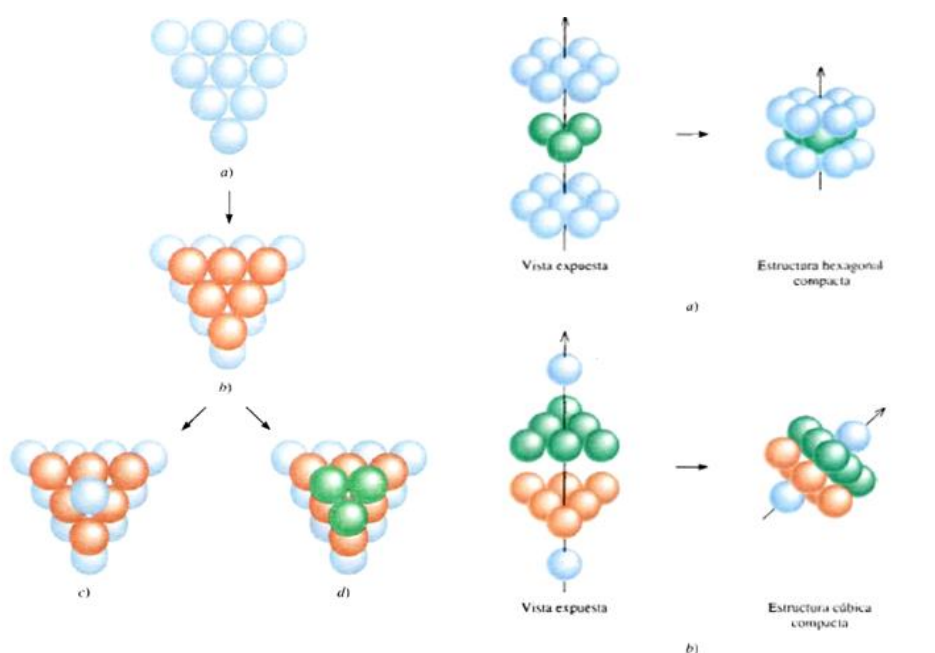


Figure 6.- Construction sequence of the different compact structures: c) hexagonal close packing and d) cubic face-centered.

Interstitial sites

In any of the crystalline structures described above, there are holes between the atoms of the lattice in which you can place smaller atoms. These spaces are known as interstitial sites.

In any compact structure a sphere rests on other three, so that the centers of the four spheres of this grouping coincide with the vertices of a tetrahedron. The space located in the centre of the tetrahedron is called tetrahedral site or hole. In any close packing lattice each sphere is in contact with three of the top layer, and with three of the bottom layer; therefore, there are two tetrahedral sites associated with each sphere (atoms).

In close packing structures there is a second type of interstitial site. This site is surrounded by six spheres, whose centers are at the vertices of a regular octahedron, and is called octahedral site or hole. An octahedral site can be generated by two sets of three spheres forming each one, in parallel planes, equilateral triangles with their vertices pointing in opposite directions. Interstitial positions in compact structures are shown in Figure 7.

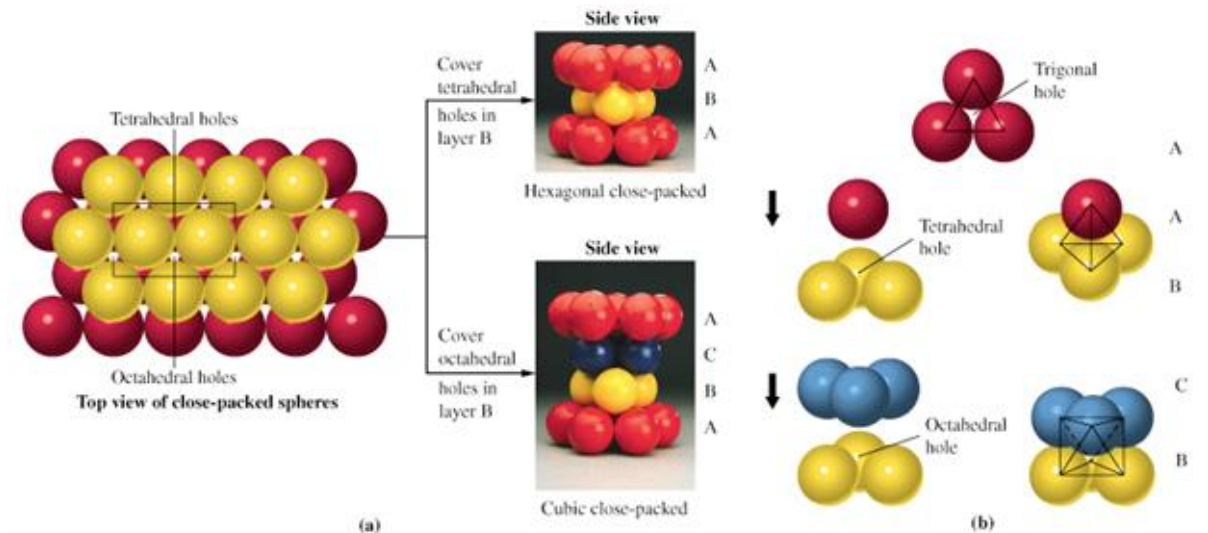


Figure 7.- Location of interstitial sites.

X-ray diffraction. Analysis of crystalline structures

A way of studying the structure of crystalline solids makes use of X-ray diffraction techniques. This is possible because the wavelength of the radiation is in the order of the interplanar spacing in crystalline solids.

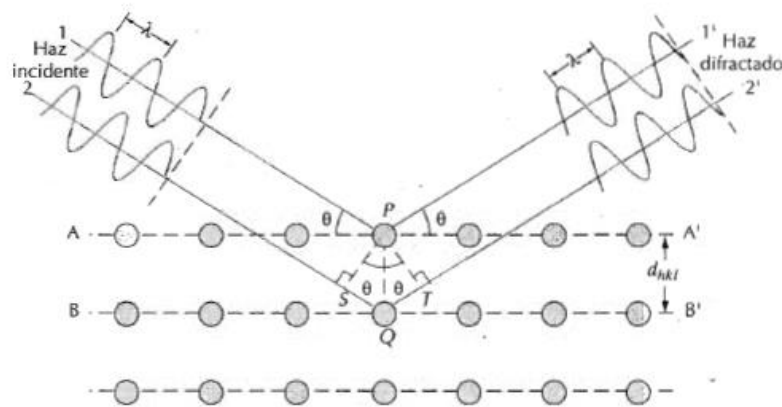


Figure 8.- Schematic representation of diffraction process.

When a beam of X-rays of known wavelength impacts on a sample, radiation scatters by the interaction with the electronic clouds of the crystal atoms. As there are many parallel planes involved in the scattering, reflections from these planes interfere with each other. Interference will only be constructive when the path-length difference between rays from successive planes equals an integer number of wavelengths. This is illustrated in Figure 8, where the rays at a particular wavelength, λ , impact at an angle θ , on a set of planes with spacing d_{hkl} . The X-ray beam diffracts at an angle given by Bragg's law:

$$2(d) \sin\theta = n \lambda \quad \text{being } n = 1, 2, 3, \dots$$

In an X-ray diffraction experiment we can get a set of reflections, resulting from the constructive interferences, that contains information about the crystallographic planes of the material. From this set we can know the way the constituents of a crystal material are ordered. The intensity of the beam scattered by a given set of planes depends on the nature of the atoms defining the planes. From the measurement of the reflections intensities it is therefore possible to identify the different atoms of the crystal.

Currently, the result of an X-ray diffraction experiment is a graph where intensity of the diffracted ray (y-coordinates) is plotted as a function of diffraction angle θ (usually 2θ) (x-coordinates). This representation is called diffraction pattern or diffraction diagram. Thus, knowing the wavelength of the X-rays and measuring the diffraction angles θ , we can calculate the characteristic interplanar spacing of a crystal. The information that can be obtained from the diffraction pattern is very broad; we are going to limit us to the calculation of lattice parameters and to the identification of substances taking already known diffraction patterns.

Reading of a diffraction pattern

The 2θ angle (x-coordinate axis) of every reflection as well as its absolute intensity (height) are measured with a graduated ruler, annotating both values in the two first columns of a five-column table. In the third column, annotate the values of relative intensities obtained as a percentage (to 100) of the most intense one. In the fourth column, calculate the d_{hkl} spacing values using the Bragg's expression ($2 d \sin\theta = n \lambda$; being $n = 1$), knowing that the $K\alpha$ radiation of Cu ($\lambda = 1.5405 \text{ \AA}$) has been used for obtaining the diffraction pattern.

With spacing values of the three most intense reflections consult the ASTM cards, selecting the one that fits better our substance problem; confirm with all the other measured reflections. The ASTM cards consist of the following areas (Table 1): interplanar spacing, relative intensity of the reflections and Miller indices, which are the characteristic parameters of the crystalline solid.

Table 1.- Areas of a ASTM card.

ASTM card number		Miller indices			
Formula	d, Å	Relative Intensity	h	k	l
Sistematic name					
Instrumental conditions for the diffraction pattern: wavelentgh used, filters, etc					
Research group					
Team data					
Crystal lattice parameters: a, b, c, α , β , γ .					
Symmetry and spatial group					
Colour:					
Preparation method					

Once the substance problem and its ASTM card are known, you can place in the fifth column of Table 2 or Table 3, depending on the symmetry, the (hkl) plane values that have led to the diffraction maxima. Since we are dealing with cubic or hexagonal cells, with the spacing values of the most intense reflections, and its (hkl) values you can calculate the cell average parameter using the expression contained in the outline, which depends on the symmetry.

For example, if it is cubic symmetry, the expression that should be used is:

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2}$$

Table 2.- Example for the diffraction pattern reading when there is cubic symmetry.

$2\theta_{\text{experimental}}$	Absolute Intensity	Relative Intensity	d (Å)	hkl	a(Å)	Crystalline solid

AVERAGE LATTICE PARAMETER:

Table 3.- Example for the diffraction pattern reading when there is hexagonal symmetry.

$2\theta_{\text{experimental}}$	Absolute Intensity	Relative intensity	d (Å)	hkl	a(Å)	c(Å)	Crystalline solid

AVERAGE LATTICE PARAMETER:

Once the “a” cell parameter is known, the unit cell volume can be calculated:

Cubic cell

$$V = a^3$$

Tetragonal cell

$$V = a^2 c$$

Hexagonal cell

$$V = \frac{(\sqrt{3}a^2 c)}{2} = 0.866a^2 c$$

The space fraction occupied by atoms, assuming atoms as rigid spheres, is known as packing factor and is calculated by:

$$\text{Packing Factor} = \frac{\sum_{\text{atoms}} \text{vol atoms unit cell}}{\text{volume unit cell}}$$

The theoretical density of a metal can be calculated using the crystalline structure properties in the following way:

$$\text{Density} = \frac{\sum_{\text{atoms}} \text{atoms unit cell} \times M}{\text{volume unit cell} \times N_{\text{Avogadro}}}$$

Where, M is the atomic mass of each atom.

EXPERIMENTAL METHOD

a) Crystalline structures using rigid spheres.

a.1.- With the help of Figures 4 and 5 and making use of 31 small spheres, build the following packings:

(i) face-centered cubic and simple cubic.

(ii) hexagonal close packing and non compact hexagonal .

a.2.- Draw the unit cell of each built structure.

a.3- Search and draw the octahedral and tetrahedral sites of the close packed structures (hexagonal or cubic).

b) From the corresponding diffraction pattern set:

b.1-Pure substance: metal

b.1.1- Index the diffraction pattern and identify what substance is using the enclosed ASTM cards. Say to which crystallographic plane corresponds each reflection in the diffraction pattern and mark them on the diffraction pattern.

b.1.2.- In the case of the diffraction pattern of a metal, calculate the unit cell parameters and once its symmetry is known, calculate the ion radius. In addition, determine the volume, the packing factor and the unit cell density. Draw in the unit cell the four families of planes whose intensities are maximum.

b.2.- Mixture of two metals:

b.2.1- Index the diffraction pattern and using the enclosed ASTM cards identify which metals are; calculate also the unit cell parameters for each metal.

ASTM cards

04-0836					
Cu Copper	d, Å	int	h	k	l
Rad: CuK _{α1} =1.5405 Filter: Ni Beta	2.0880	100	1	1	1
Cut off: Int: Diffract. I/Icor.:	1.8080	46	2	0	0
Ref: Swanson, Tatge, etc (1953)	1.2780	20	2	2	0
Sys.: Cubica (FCC) SG.: Fm3m (225)	1.0900	17	3	1	1
a: b: c: A: C:	1.0436	5	2	2	2
α: β: γ: Z: 4 mp: 1083	0.9038	3	4	0	0
Dx: 8.935 Dm: 8.950 SS/FOM	0.8293	9	3	3	1
Colour: Red	0.8083	8	4	2	0
Preparation method					

04-0787					
Al Aluminium	d, Å	int	h	k	l
Rad: CuK _{α1} =1.5405 Filter: Ni Beta	2.3380	100	1	1	1
Cut off: Int: Diffract. I/Icor.:	2.0240	47	2	0	0
Ref: Swanson, et al., Natl. Bur. Stand. (U.S.), Cir. (1953)	1.4310	22	2	2	0
Sys.: Cubic (FCC) SG.: Fm3m (225)	1.2210	24	3	1	1
a: b: c: A: 4 C:	1.1690	7	2	2	2
α: β: γ: Z: mp:	1.0124	2	4	0	0
Dx: 2.699 Dm: SS/FOM	0.9289	8	3	3	1
Colour: Metallic grey	0.9055	8	4	2	0
Preparation method	0.8266	8	4	2	0

06-0696					
Fe Iron	d, Å	int	h	k	l
Rad: CuK _{α1} =1.5405 Filter: Ni Beta	2.0268	100	1	1	0
Cut off: Int: Diffract. I/Icor.:	1.4332	20	2	0	0
Ref: Swanson, et al., Natl. Bur. Stand. (U.S.), Cir. (1955)	1.1702	30	2	1	1
Sys.: Cubic (BCC) SG.: Im3m (229)	1.0134	10	2	2	0
a: b: c: A: C:	0.9064	12	3	1	0
α: β: γ: Z: 2 mp:	0.8275	6	2	2	2
Dx: 7.875 Dm: SS/FOM					
Metallic grey					
Preparation method					

04-0831					
Zn Zinc	d, Å	int	h	k	l
Rad: CuK _{α1} =1.5405 Filter: Ni Beta	2.4730	53	0	0	2
Cut off: Int: Diffract. I/Icor.:	2.3080	40	1	0	0
Ref: Swanson, et al., Natl. Bur. Stand. (U.S.), Cir. (1953)	2.0910	100	1	0	1
Sys.: Hexagonal (HCP) SG.: P6 ₃ /mmc (194)	1.6870	28	1	0	2
a: b: c: A: C:	1.3420	25	1	0	3
α: β: γ: Z: mp:	1.3320	21	1	1	0
Dx: 7.136 Dm: 7.050 SS/FOM	1.2370	2	0	0	4
Colour: White	1.1729	23	1	1	2
Preparation method	1.1580	5	2	0	0
	1.1236	17	2	0	1
	1.0901	3	1	0	4

SESSION II: MECHANICAL PROPERTIES **(TENSILE TEST)**

AIM

To determine important aspects of strength and elongation of materials that can be used for quality control, material specifications, and the calculation of parts subjected to stress.

MATERIAL

- Aluminum samples.
- Polymeric samples
- Composites samples (From session II)
- Gloves, tweezers, and protection elements
- Caliber
- Tensile test equipment
- Calculator
- Ruler

INTRODUCTION

The stress-strain mechanical test is the most common performed in tension. The tensile test can be used to determine various properties of materials. Normally, a specimen is deformed to rupture, when an increasing tensile load is applied gradually and uniaxially along the axis of the specimen.

Tensile tests are performed on materials such as metals, thermoplastics, thermostable plastics reinforced with glass fiber, as well as phenolic resin molded or laminated composite materials.

There are different standards for the tensile test, DIN 53455, ISO / DP 527, ASTM 638. We will use ASTM.

Test samples

The test specimens for metallic materials are generally obtained by machining of a sample of the product under test, or a molded sample. In the case of products with a constant section (profiles, rods, etc.) or obtained by molding bar they can be used as samples without machined specimens. The section of the specimen may be circular, square or rectangular.

Usually the test specimens for non-metallic materials can be prepared by compression, injection or by metal removal by cutting plates. There are generally three types of probe:

a) Semi-rigid and rigid plastics

The specimens will be formed according to the dimensions in Figure 1. The sample type MI is the preferred sample and it will be used when laying a sufficient material thickness of 10 mm or less.

The type of specimen M-III will be used when the material being tested has a thickness of 4 mm or less and the type of specimen M-II is used when required direct comparisons between materials with different stiffness (non-rigid and semi-rigid).

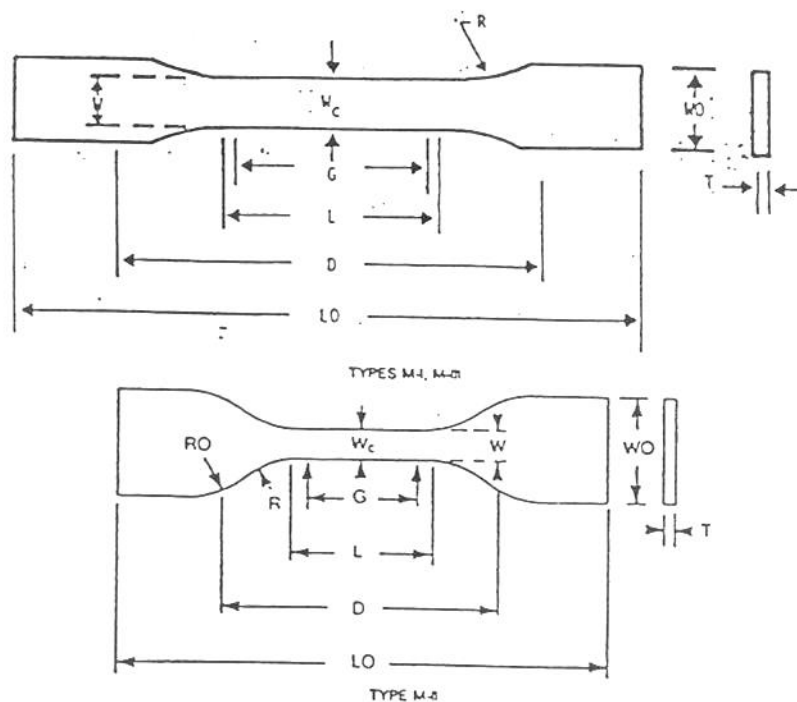


Figure 1.- Types of specimens used in tensile tests on plastics.

b) Non-rigid plastic

It uses the type of specimen M-II with a thickness of 4 mm or less. MI type specimen should be used for all materials with thicknesses between 4 and 10 mm.

c) Reinforced composite materials

The specimens of reinforced composite materials are of the type MI. In all cases the maximum thickness of the specimen is 10 mm. The specimens to be tested must be submitted free surface defects, scratches, or imperfections. The marks for machining operations of the specimen will be carefully removed. The final finish will be in a direction parallel to the long axis of the specimen.

If necessary make marks for grips, they are made with crayons and India ink, which do not affect the material. Never make scratches or marks with punches. When you suspect the presence of anisotropy in mechanical properties they will be samples with identical dimensions having their long axes parallel and perpendicular to the direction of anisotropy suspected.

How tensile test works

The test consists in deforming a specimen uniaxial stretch and record this deformation *versus* the force applied. It is done in dynamometers or machines with adjustable speed drive and a graphic record. The obtained diagrams, called stress-strain diagram, have the form shown in Figure 2. In this figure it is shown the stress-strain curve and the various parameters that can be obtained from it.

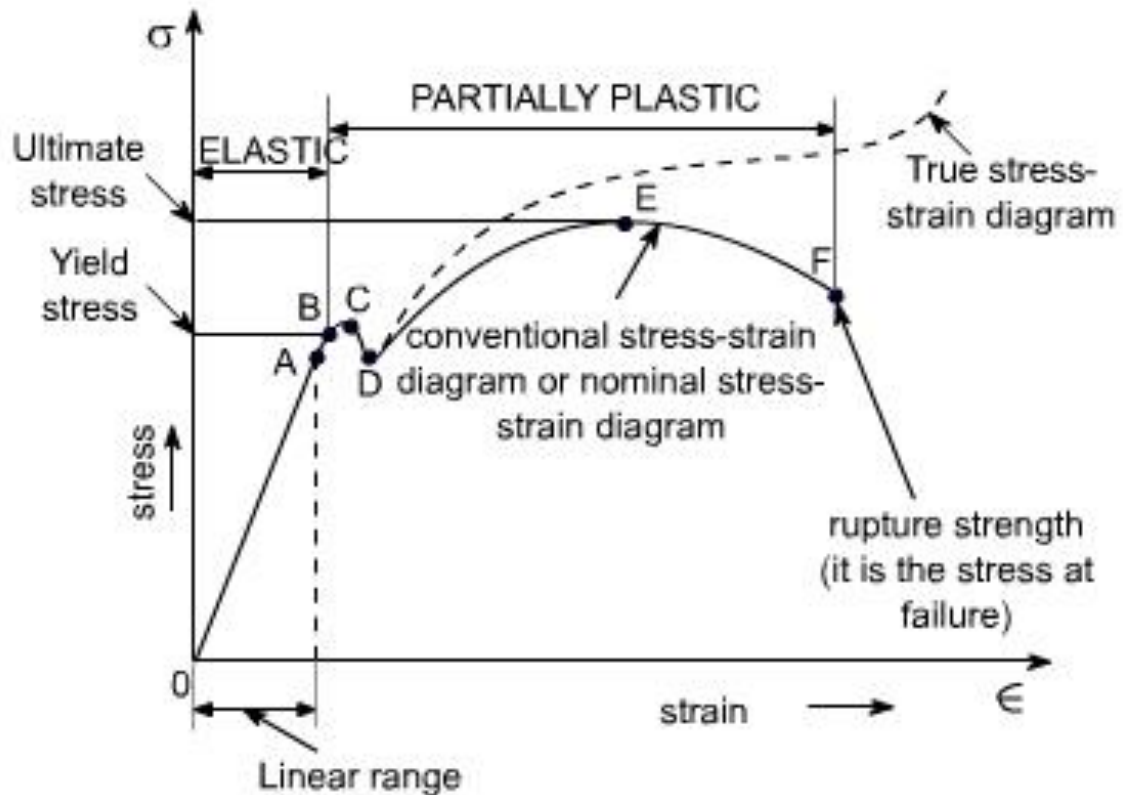


Figure 2.- Typical stress-strain curve.

Stress-strain curves generally show a linear region at low loadings, and a line starts at 0.002 strain and tangent to that portion of the curve permit to calculate the modulus of elasticity $\sigma_y = \sigma (0,2\% \text{ of } \epsilon)$.

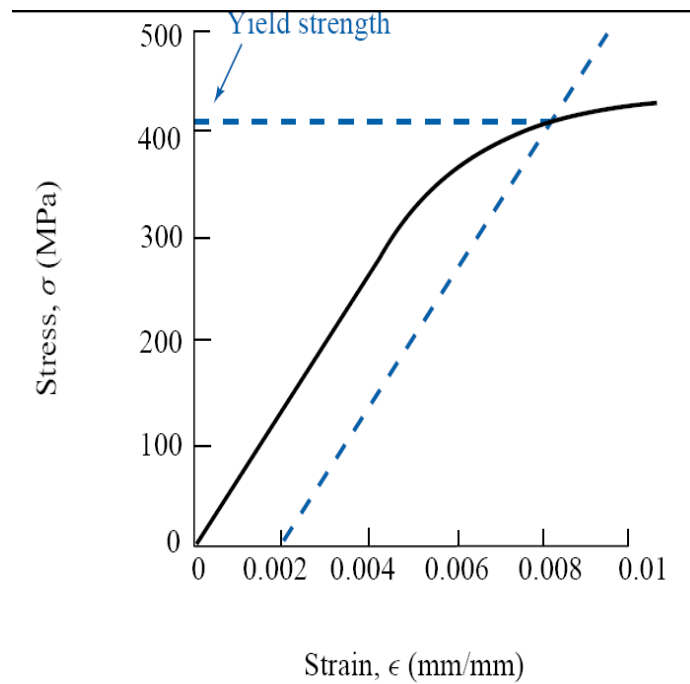


Figure 3.- Conventional yield stress graphic calculation(σ_y).

The sample is placed inside the tensioning grips, so that fit well and have wedge effect with pneumatic, hydraulic, or manual. The initial force should not be too high, otherwise it could distort the test result. Also, it is important take care that there is no slippage of the specimen. The testing machine is designed to extend the specimen at a constant speed and continuously and simultaneously measuring the instantaneous load applied (with a load cell) and the resulting strain (using an extensometer). The test takes a several minutes and is destructive. During the test specimen it is deformed and often permanently broken. The rate of stretch is following the ASTM standard.

Factors affecting tensile test

As it is known, the mechanical properties of materials depend mainly on their composition and structure, but there are other factors that influence and are mainly the same, time or speed of application of load, temperature, humidity, history previous sample, etc. Consequently, where precise comparative results are required, these factors must be carefully controlled. Also, when it comes to comparing results all samples should be prepared in the same way.

OPERATING PROCESS

- 1) Measure the length, the width, and thickness of the specimens with a vernier or caliper at different points along the sample section.
- 2) Make a mark on the specimen to measure the maximum elongation subsequently experienced.
- 3) Place the specimen in the testing machine and secure it with the grips.
- 4) Select the speed test according to ASTM standards. Always must be the one that causes rupture of the specimen in a time between 0.5 and 5 minutes.
- 5) Place the two parts of the tested sample back together and measure the new length.
- 6) Take note of all data and observations in your report Table 1.

SESSION III- IDENTIFICATION OF POLYMERS. **ANALYTICAL PROCEDURES.**

AIM

Perform preliminary tests to identify analytical type of polymers used industrially.

MATERIAL

- Several samples of polymers
- Very sharp razor
- A test tube
- Clamp
- Beaker
- Wash bottle with distilled water
- Universal indicator (paper)
- Goggles
- Bunsen burner of adjustable flame
- Two spatulas of annealed nickel
- Dishes of porcelain or metal
- Aluminum foil or sheet steel
- A pure copper wire with a thickness of 2 mm and 20 mm in length
- Vials of 100 ml glass-stoppered
- 1 liter of the following solvents: toluene, chloroform (CH_3Cl), ethyl ether, ethanol, tetrahydrofuran (THF), aqueous NaOH, and H_2SO_4 aqueous solution

INTRODUCTION

The term polymer includes all materials based on chemical compounds formed by long chains composed of units, which are low molecular weight molecules (monomers), linked covalently. Each monomer has a different physical and chemical property. The formed polymers keep part of those properties and also, by its own structure, they add new ones. Very different polymeric materials properties can be obtained combining two or more monomers, depending on the relative amount of each. The polymer chains can be linear or branched, be united to form a crosslinked structure or not. Polymers may have a range of critical applications by combining the possibilities in the composition and structure. In general, polymers are poor conductors of heat and electricity because, apart from the directional nature of covalent bonds, have no free electrons. In addition, comparatively to other materials they have lower density due its structure is relatively less compact. Polymers can replace other materials which present similar performance or they may be used to save other technical problems, such as those in spring joints, hinges film, structural foams, non-lubricated bearings, special sliding, etc. Together with ceramics,

polymers can compete with traditional metals in terms of technical properties and outperform them in terms of corrosion resistance and cost.

The plastic, as an industrial product, is a combination of polymers and a variable number of other substances, fillers, additives, which are added to give the final material characteristics or specific features. In some cases, polymer may need to be treated prior to the analysis, in which polymer additives and its low molecular weight impurities must be separated. The identification process consists of submitting the material to a series of preliminary tests such as solubility, density, melting, and smell of the flame. Sometimes, it is necessary to perform specific tests such as identifying the presence of heteroatoms.

CHARACTERISTICS OF POLYMERS

These materials have very different characteristics compared to the organic substances of low molecular weight. They are solids at room temperature, viscous, elastic or brittle, and some of them may form films or fibers. The macromolecules can be of several types:

- a) Linear, composed of monomers that have only two binding sites.
- b) Branched when its constituent monomers can be linked by three or more places with its neighbors.
- c) Crosslinked macromolecules chains and branches intertwined in the three directions of space.

SIZE AND SHAPE OF POLYMERS

A polymer is formed by a set of macromolecules whose size is irregular. The size of the macromolecules is determined by their molecular weight. These materials do not present a specific molecular weight but they exhibit a molecular weight distribution. The determination of the molecular weight distribution should be done according to statistical averages:

Average number or first moment of distribution:

$$\overline{M}_n = \frac{\sum N M_i}{\sum N}$$

Weight average or second statistical moment:

$$\overline{M}_w = \frac{\sum W M_i}{\sum W}$$

The width of the molecular size distribution can be calculated from the polydispersity index, defined by: $PD = M_w/M_n$.

The size of the polymer significantly affects the mechanical and thermal properties.

Thus, glass transition temperature of the polymer (T_g) varies with molecular weight

according to: $T_g = T_g^\infty - \frac{k}{M_n}$

The spatial arrangement of groups and substituents with respect to the skeleton of the macromolecule is important because they influence on the intermolecular cohesive forces and hence the crystallinity and physical properties. Depending on the spatial arrangement of groups, polymers can be classified into:

- Isotactic: everyone on the same side
- Atactic: groups randomly distributed on each side
- Syndiotactic: groups alternating on each side

CRYSTALLINITY

Almost all solids tend to be crystalline due to the regular arrangement of the particles that form them. In the case of polymers this sort is difficult due to the complexity of the macromolecules. Crystalline polymers tend to form in certain areas of the solid clusters similar to crystals, which are called spherulites. The crystalline regions of a polymer do not have the regular form of regular crystals. They are much smaller, contain many imperfections, and are connected through amorphous areas. Those regions are shared by a chain and another area, so there is no clear separation between amorphous and crystalline areas.

Factors contributing to the ordination process are:

- a) Regularity in the structure.
- b) Freedom of rotation of the macromolecule chains.
- c) Presence and regular distribution of groups capable of forming intermolecular bonds.
- d) Absence of bulky groups.

A polymer with more or less crystallinity has a great importance in mechanical and thermal properties of the material

SOLUBILITY

The polymers have a dissolution process different from that observed in molecular weight compounds. The reason for this behavior is directly related to the size and structure of the polymer. A dissolution process involves the step of solvent molecules between the molecules of solute which gives rise to intermolecular interactions solute-solvent and weakening interactions solute-solute or solvent-solvent.

For a polymer, this process is hampered by the entanglement between the chains. With increasing branching or crosslinking of the polymer solvent molecules present more difficult to penetrate. A first stage in which the solvent enables the elimination of entanglements must be done and then the polymer will be swell. Once this stage is finished, common to all polymers, more solvent molecules could solvate the polymer chains to form a single phase.

Table 1.- Solvents and solvent for most common polymers.

Polymer	Good Solvent	Poor solvent
Polyethylene, polypropylene atactic and isotactic	p-Xylene, trichlorobenzene Hydrocarbons (> 80 ° C)	Acetone, diethyl ether, methanol, ethanol, ethyl acetate.
Polybutadiene, polyisoprene	Aliphatic and aromatic hydrocarbons	Acetone, diethyl ether, methanol, ethanol
Polystyrene	Benzene, toluene, chloroform, cyclohexanone, THF, methyl ketone, DMF	Ethanol, methanol, diethyl ether (swollen)
Polyvinyl chloride	THF, cyclohexanone, methyl ketone, DMF	Methanol, acetone, heptane
Vinyl polyacetate	Benzene, chloroform, methanol, acetone	Diethyl ether, butanol
Polytetrafluoroethylene	Insoluble	Insoluble
Polyacrylates and polymethacrylates	Chloroform, acetone, THF, DMF, toluene, ethyl acetate	Methanol, diethyl ether

Polyacrylonitrile	DMF, dimethyl sulfoxide, concentrated sulfuric acid	Alcohols, diethyl ether, water, hydrocarbons
Polyacrylamide	Water	Methanol, acetone
Cellulose	Cupric hydroxide, aqueous ammonia, aqueous zinc chloride, calcium thiocyanate aqueous	Methanol, acetone
Aliphatic polyesters	Chloroform, formic acid and benzene	Methanol, diethyl ether, aliphatic hydrocarbons
Polyamides	Formic acid, concentrated sulfuric acid, DMF	Methanol, diethyl ether, hydrocarbons
Methylene polyoxides	DMF, benzyl alcohol	Methanol, diethyl ether, hydrocarbons
Polydimethyl siloxane	Chloroform, heptane, benzene, diethyl	Methanol, ethanol
Epoxy resins	Insoluble	Insoluble

DENSITY

The density of a material (ρ) is the ratio between mass (m) and volume (V): $\rho = m/V$. A simple method to determine the density of the polymer is by flotation. This approach implies that the sample of the polymer floats in a liquid of known density. For example, samples can be dipped in methanol ($\rho = 0.79 \text{ g cm}^{-3}$), water ($\rho = 1.00 \text{ g cm}^{-3}$), etc. Behavior of the sample determines whether the polymer has lower or higher density than the liquid in which it is immersed.

Table 2.- Average density of some polymers.

Polymer	Density (g cm^{-3})	Polymer	Density (g cm^{-3})
Silicone	0.80	PP	0.85-0.92
LDPE	0.89-0.93	natural rubber	0.94
HDPE	0.94-0.98	Nylon 12	1.01-1.04
PS	1.04-1.06	ABS	1.04-1.08
Nylon 6	1.07-1.09	Polyester resin	1.10-1.40
PAN	1.14-1.17	PVA	1.17-1.20
PMMA	1.16-1.20	PC (polycarbonate)	1.20-1.22
PU (polyurethane)	1.20-1.26	PEEK (polyetheretherketone)	1.26-1.32
PEI (polyetherimide)	1.27	PET (polyethylene terephthalate)	1.38-1.41
PVC	1.38-1.41	PTFE	2.10-2.30

MECHANICAL BEHAVIOR

The singular constitution of plastics and the nature of the binding forces explain why they have a less compact structure than the dense packing of metallic crystals. Hence, they exhibit usually a lower strength, lower modulus, and sensitivity to the impact.

Table 3. - Tensile strength of some polymers in pounds / inch² (psi).

Polymer	Tensile strength (psi)
Low Density Polyethylene	1000-2400
Polytetrafluoroethylene	3500
High Density Polyethylene	4400
Polydimethylsiloxane	5000
Polypropylene	5000
Polystyrene	8000
Polyamides	9000-12000
Polycarbonate	9500
Polyesters	10000
Polysulfones	10200-12000
Epoxy resins	16000

As a result of the different phenomena involved, the mechanical behavior of a polymer is very complex and therefore depends on a number of factors:

- Crystallinity and mobility of the chains
- Molecular weight
- Resistance to deformation
- Working temperature
- Speed and duration of application of stress (direct consequence of the viscoelastic nature of these materials).

A polymeric material subjected to tensile or uniaxial tension experiences a resistance to elongation or break that depends heavily on temperature and test speed. If we consider a tensile test at constant temperature, stress-strain curve shows a number of zones corresponding to different behaviors of the material. The behavior in traction for a non-crosslinked polymeric material corresponds to one of the following five types:

a) Soft and weak have a low tensile strength and moderate elongation. It is the typical behavior of non-crosslinked elastomers.

b) Soft and tough exhibit large elongation with moderate tensile strength. It is present in rubber and polyvinyl chloride (PVC) plastic.

c) Hard and brittle supports low elongation with high tensile strength. Examples are polystyrene (PS) and polymethylmethacrylate (PMMA).

d) Hard and strong they have a high tensile strength with high elongation. It is the case of polyvinyl chloride (PVC).

e) Hard and tough present yield point, a high elongation and good tensile strength. This behavior is present in polyamides, nylon, polyesters, and cellulose derivatives.

THERMAL BEHAVIOR

The thermal behavior of polymers is characterized by melting temperature (T_m) and glass transition temperature (T_g) that define their properties and resistance. If the polymer is amorphous behaves as a glass at low temperature and become gummy above its T_g , and far above, flows like viscous liquids. Crosslinked amorphous polymers do not flow and remain as gums. If the polymer is crystalline, melts upon reaching its T_m , from a viscous liquid. A semi-crystalline polymer has a combined behavior, becoming something rubbery above its

T_g and melting after reaching its T_m . In general, the value of T_m is 30 to 100% higher than T_g .

The polymers, whose chains are very flexible structures, have very low values of T_g . The interactions between side groups of polymer chains make T_g increase. Thus, the polyvinyl chloride has a T_g greater than that of polyethylene due to dipole-dipole interactions between the chains of chlorine. Groups that produce rigidity, such as aromatic rings with different substituents, make both T_g and T_m of polymers with these rigid structures are higher than those of similar polymers. If aliphatic groups $(-CH_2)_n$ are present in the chains, they make them much more flexible. Thus, polystyrene (PS) has a much higher T_g than polyethylene (PE). However, the PE is not softened before the PS, because it is amorphous and the only temperature that leads their behavior is T_g , while the PE has a very high degree of crystallinity and the temperature that determines its properties is especially the T_m which results to be greater than the T_g of PS.

a) Thermoplastic polymers

They consist of linear or slightly branched macromolecules and behave in a plastic and ductile manner. At elevated temperatures, these polymers are soft and comprised of viscous flow, while maintaining its shape after cooling. In addition, thermoplastic polymers can be reheated and shaped again without significantly changing its properties.

Table 4. - Properties of some thermoplastic polymers.

Name	Registered Name	Properties	Applications	Formula
Polyethylene (PE)	Polythene, Rigidez, Alkathene, Hostales, Lupolen, Alathon	Cheap, easy processing, good chemical resistance, poor resistance under severe environmental conditions	Household utensils, packaging, containers	$-(CH_2-CH_2)-$
Polypropylene (PP)	Propathene, Appryl, Novelen	Cheap, easy processing, good chemical resistance, poor UV resistance	Packaging, containers, furniture, pipes	$* \left(CH_2 - \underset{\substack{ \\ CH_3}}{CH} \right)_n *$
Polyvinyl chloride (PVC)	Darvic, Corvic, Geon, Evipol, Vinnolit, Hostalit	Inexpensive, rigid, Good chemical resistance, limited thermal stability, additives required for processing	Packaging, cable insulation, pipes, toys	$* \left(CH_2 - \underset{\substack{ \\ Cl}}{CH} \right)_n *$
Polystyrene (PS)	Styron, Polystyrol, Novacor	Cheap, transparent, rigid, good insulating properties, low water absorption, brittle	Good containers, packaging, appliances	$* \left(CH_2 - \underset{\substack{ \\ \text{C}_6\text{H}_5}}{CH} \right)_n *$
Polymethylmethacrylate (PMMA)	Perspex, Plexiglas, Lucite, Acrylate	Transparent, good environmental properties, tough, stiff, poor insulating properties, poor resistance to organic solvents	Transparent sheets, airplane windows, lamps, displays	$* \left(CH_2 - \underset{\substack{ \\ O=C-OCH_3}}{C} \right)_n *$
Polyamide (PA)	Nylon, Ultramid, Zytel, Caprolan, Stanyl, Capron, Akulon, Rilsan, Vestamid	Tough, flexible, abrasion resistance, good friction and wear properties, absorbs water	Textile, brushes, surgical appliances, clothes, ropes	$-[NH-(CH_2)_5-CO]-$

Polytetrafluoroethylene (PTFE)	Teflón, Fluon, Halon, Hostaflon	Low friction, good electrical insulation, excellent chemical resistance, insoluble, relatively expensive	Nonstick surfaces, duct tape, engineering applications	$-(CF_2-CF_2)-$
Polyacrylonitrile (PAN)	Barex, Orlon	Strong, good chemical resistance	Applications similar to wool, membranes	$-\left[CH_2-\underset{\substack{ \\ C \equiv N}}{CH}\right]_n$
Polyvinylacetate (PVA)	Elvacet, Vinylite	Good overall stability, quick drying, inexpensive	Surface coatings, adhesives, paints	$-\left[CH_2-\underset{\substack{ \\ O \\ \\ C=O \\ \\ CH_3}}{CH}\right]_n$

b) Thermosetting Polymers

Thermosetting polymers have a crosslinked structure, which can be formed by heating or chemical reaction. They have excellent thermal stability and rigidity. It does not melt and can only be formed during the synthesis process.

Table 5. - Some properties of thermoset polymers.

Name	Registered Name	Properties	Applications
Phenolic resins	Bakelita, Plenco, Durite	Good chemical resistance, thermal and flame	Electrical appliances, fixtures and furniture, adhesives, laminates
Epoxy resins	Araldite, Epon, Epikote	Tough, good adhesion, excellent chemical resistance, good electrical properties	Adhesives, coatings, laminates, construction, electrical and electronic components
Amino resins	Avisco, Plaskon	Rigid, strong, good impact resistance	Kitchen utensils, flooring, plywood
Polyester Resins	Laminac, Aripo, Baygal	Low viscosity, cheap	Automotive engineering, construcción, boats
Polyurethane	Lycra, Elastane	Burn easily produced smoke, good chemical resistance and abrasion, high elasticity, excellent adhesion properties and impact	Furniture, tires, clothing, kitchen surfaces and floors.

c) Elastomers

Elastomers or rubbers are polymers that by applying a tension they can deform easily, but with the particularity that removing the effort, the polymer returns to its original shape. This property which characterizes the elastomer is a result of weakly crosslinked structure, forming a lattice with a low crosslinking density. The number of binding sites depends on the number of functional groups of the starting monomer and it affects elasticity.

Table 6.- Properties and applications of some elastomers.

Name	Registered Name	Properties	Applications
Poly (cis-isoprene) (natural rubber)	Natsyn, Amenpol	Good abrasion resistance, poor heat resistance, good electrical properties	Gaskets, shoe soles, condoms
Silicone	Silactic, Rimplast, Sinale	Thermally stable, resistant to water, expensive	Medical implants, sealants, flexible molds, electrical boards
Polychloroprene	Neoprene, Perbunan	Resistant to oils, good resistance to environmental conditions, low flammability	Cables, gaskets, hoses, seals

PERFORMANCE HEAT

Heating a polymer sample may provide important information into the nature of the polymer. The tests performed are two:

- a) **Flame behavior:** a small sample is maintained with a spatula or metal tweezers, in the lower part of the flame of a Bunsen burner. The appearance of the flame and the production of odor are characteristic for each type of polymer.

Table 7.-Behavior of polymers to the flame and characteristic odor.

Polymer	Inflammability /Combustión	Flame appearance	Smell
PTFE	Burn very little, virtually fireproof	Yellow	It does not smell
PET		Yellow, gray	Sweet
PE	It is moderately burn, Readily combustible, Easily ignited droplets merge	Yellow, blue base	Burning wax
PP		Yellow, blue base	Burning wax
PVC	Difficult to ignite, self-extinguishing by removing the flame, moderately combustible	Yellow, green base	Acre, hot, irritable
Nylon		Yellow, blue smoke	Horn burned
PC		Bright yellow, black smoke	Phenols
PS	It is easily ignited, continues to remove the flame burning	Yellow, blue base, light, black smoke	Similar natural gas, something fragrant, styrene
PMMA		Yellow, blue base, bright	Sweet, fruity
Rubber		Yellow / orange, dark, black smoke	Burning rubber
PVA	The flame burns, not self-extinguishes or it is possible when the flame is removed slowly	Yellow	Acetic acid
Epoxy resin	Difficult to ignite	Orange / yellow, gray	Acre, spicy
Polyester resin	Burn moderately	Yellow, blue base, medium, black smoke	Styrene gas
PU	Moderately combustible	Yellow, blue base, white smoke	Acre, hot, irritable

- b) **Pyrolysis test:** with this test we can study the behavior of polymer degradation. Thermal degradation produces low molecular weight fragments that are often flammable or have distinctive odors. The polymer sample is introduced into a pyrolysis tube and held with tweezers to the Bunsen burner flame. At the top stands a pH indicator paper to determine the nature of the smoke. The polymer can be distinguished as acidic, basic or neutral.

Table 8.- pH range slowly heated polymer vapor tube.

pH 0.5-4.0	pH 5.0-5.5	pH 8.0-9.5
Polyester resin	Silicone	Nylon
PU elastomer	PP	PAN
Cellulose acetate	PE	Amino resin
Cellulose nitrate	PS	Phenolic resin
PET	Epoxy resin	ABS
PVC	PVA	
PTFE	PMMA	
	PC	
	PU crosslinked	

OPERATING PROCESS

The practice will take place in groups of two or three people and each group will have several samples of polymers. By performing all identification tests described below, groups will be able to identify the polymer.

PRELIMINARY TESTS

The following characteristics to be observed are a function of material thickness and the type and percentage of filler loading.

Visually one can distinguish amorphous thermoplastics, because, in general, are transparent or translucent as glass. The assessment is always easier in a natural state and without load. The semi-crystalline thermoplastics are opaque (milky turbidity).

From the **surface** appearance, it should be noted that parts made with amorphous and semi-crystalline thermoplastics have a surface partially shining, this is observed in the samples in sheet form. In this part of the practice is only necessary to observe and record the external aspects present in the samples, without any manipulation.

MECHANICAL BEHAVIOR

Thermosets are basically more brittle than thermoplastics. The experimental part of this section consists on observing the behavior after cutting with a knife (use a razor blade). It can easily distinguish between thermoplastics and thermosets. The procedure is as follows: Make a cut with a very sharp razor and slide it beneath the surface trying to start shaving. Thermoplastic polymers are chipped including those that are brittle and reinforced with fiberglass. Thermosets are not chipped, sometimes they just are shredded fragments.

THERMAL BEHAVIOR

The test should be performed with great conscientiousness and, if necessary, be repeated.

Burning behavior

For the test you need: a test tube, clamp, beaker, wash bottle with distilled water, universal indicator (paper), and goggles.

Procedure: the sample, cut into fragments, is introduced into a test tube again and it is remained for 1 minute in the flame, on the small Bunsen burner. To analyze the vapors, a strip of indicator paper moistened with distilled water is introduced inside the tube. If necessary, heat for a further minutes until formed vapors can reach the top of the test tube.

Comments:

a. Melting behavior: the substance melts, it melts; becomes fluid, it is viscous; the melt mass becomes dark colored and bubbles appear, it decomposes.

b. Coloration of paper indicator: if it remains colorless, the reaction will be neutral; if turns red, the reaction is acidic, and if it is blue, alkaline; it should always be considered the intensity of staining.

Observe and record all changes in the polymer and the color of the indicator paper. Using the tables presented above, identify the type of polymer with which we work.

Evaluation of the combustion behavior

For the **test**, use a small Bunsen burner flame and two spatulas annealed nickel. It is useful to have dishes of porcelain or metal, even aluminum foil or sheet steel, to collect the drops of molten substances.

Procedure: the plastic is not exposed to direct flame, but kept on nickel spatula. The sample is placed on the spatula and it is heated with a small flame, in such a way it is possible to observe the behavior of combustion. When it starts to burn, the spatula is removed from the flame and we can see their post-combustion.

Comments:

a. Inflammability: plastic can be easy or difficult to ignite.

b. Combustibility: plastic burns, not burn, show dense smoke, it becomes incandescent.

c. Color flame: bright or black; may splash or drop.

d. Smell of vapors: the appreciation of odors is somewhat subjective, so it is desirable to have a collection of known smells of plastics.

e. Residue on the spatula: when the plastic is completely reduced to ashes, the calcining residue allow us to identify if the plastic was reinforced with fiberglass, glass balls, ground mineral fillers etc, by inorganic fillers. Sediment on the spatula is due to pigments and other additives are not noticeably affecting the density.

Observe and record all changes in the polymer and the color of the indicator paper. Using the tables presented above you can identify the type of polymer with which we are working.

CHLORINE TEST (Beilstein test)

This essay is based on the green color of the flame which confirms the presence of chlorine or bromine, in general, halogens.

For **testing** you will need: Bunsen burner of adjustable flame, a pure copper wire with a thickness of 2 mm, and 20 mm in length.

Procedure: the copper wire is subjected to the flame until it ceases to give color, then the polymer is touched with a glow-wire and comes back to the flame. If the material traces of the flame burns green, it indicated the presence of chlorine (or bromine).

The reaction which takes place can be represented as follows:



This test is not specific because compounds with nitrogen may obtain a flame of the same color, due to the formation of $(\text{CN})_2\text{Cu}$.

The green color of the flame is produced in PVC and derivatives (including PVC plastic), with chlorinated PE and chlorinated rubber, but also in the fireproof plastic (fire protected) with chlorinated or brominated additives, although it contains no chlorine in basic polymer.

If the flame is green, in some cases is necessary to verify if the chlorine (or bromine) is in the same molecular structure of the polymer or it is only an additive. If vapor reaction is very acidic, then it is likely that chlorine is part of the same basic molecule.

Perform the test on the polymer samples to identify which of them has halogen atoms in its structure.

SOLUBILITY

For the **test** are required vials of 100 ml glass-stoppered, 1 liter of the following solvents: toluene, chloroform (ClCH_3), ethyl ether, ethanol, tetrahydrofuran (THF), aqueous NaOH and H_2SO_4 aqueous solution. We must be very cautious in handling solvents of low boiling point; no smoking, no subjecting them to direct flame, no sparks and no discharge into the piles to flow into a drainage stream.

Procedure: For a better comparison, practice tests will be under identical conditions of temperature and time. Enter a few pieces of polymer in the vial with 50 ml of solvent; close the bottle with the cap, and leave at room temperature at least 10 h. After a while, make the following observations: *dissolved*, *swollen*, *not dissolved*. Table 1 shows the solubility of some polymers in different solvents.

DENSITY

For the **test**, take several solvents with different densities which do not solubilize the polymer. For example, methanol ($\rho = 0.79 \text{ g cm}^{-3}$), water ($\rho = 1.00 \text{ g cm}^{-3}$) and chloroform ($\rho = 1.47 \text{ g cm}^{-3}$); and polymer samples from slabs with known dimensions and mass.

Procedure: The tests should be performed at the same temperature.

Comparison method: Enter a sample of polymer in each solvent and observe if the polymer remains on the surface, floats in the liquid, or immerses. By comparison with the densities of liquids can be estimated if the density is greater or less than one. Table 2 shows the values of densities for various polymers.

Pycnometry: density determinations are performed in a pycnometer which is a container perfectly calibrated in volume at a certain temperature. Weigh the empty pycnometer (m_1), introduce a certain amount of polymer and weigh (m_2), and then fill to the mark the remaining volume with a liquid of known density (ρ) and weigh again (m_3). Remove the polymer and dilute to the mark with the liquid of known density to make the last weighing (m_4). The mass of the polymer will be $m_2 - m_1$ and volume, the fluid it displaces $(m_4 - m_3 + m_2 - m_1)/\rho$. The precision of this method reaches $\pm 0.004 \text{ g cm}^{-3}$.

SESSION IV: COMPOSITE MATERIALS

AIM

To give students a basic understanding of composite materials and their usage in design of engineering structures. The concepts will be applied to design and build a simple sandwich composite structural sample.

MATERIAL

- Prepreg composites laminates
- Scissors
- Cutter
- Gloves, tweezers and protection elements
- Caliber
- Autoclave equipment
- Calculator
- Ruler

INTRODUCTION

In a world of materials there are many choices to suit many needs: for buildings concrete, for machines steel, for packaging plastics and cardboard, etc. All of these materials have specific properties that make them useful for some applications, and not useful for others. Hence, what is done when an application exists where no one material exhibits the desired properties? Different materials with the desired properties are combined to meet this need and the resulting material is a composite. Composites are seen in many applications, from the mundane such as fiberglass used in shower enclosures and skateboards, to highly engineered composites used in today's most modern airplanes.

Composites are typically used in applications where a material of high strength and low weight is desired. For instance, while metals typically have high strength, they are among the heaviest materials. Plastics, while very light, tend to be comparatively weak. Ceramics in fiber form, while very strong, lack rigidity without additional support. Then, if these ceramic fibers are placed in plastics, the fibers may carry most of the force, while the plastic helps the fibers to maintain the desired useful shape. Figure 1 shows a simple fiber reinforced composite.

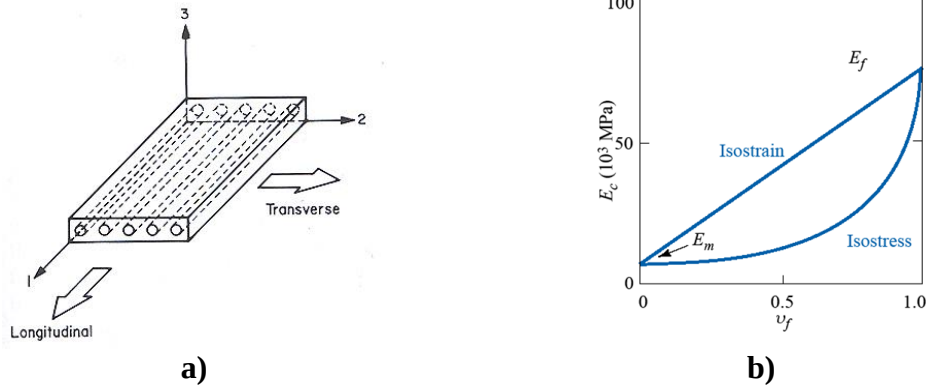


Figure 1.- a) Schematic of fiber reinforced composite lamina with loading directions indicated
b) Plot $E_c V_f$ to Isostrain and Isostress condition.

Composite Properties

The final properties of a composite are determined by fiber content, matrix material, fiber material, fiber orientation, and to a smaller extent by the fiber length and distribution in the composite. The reinforcing effects of the fiber in the composite are best explained by the rule of mixtures. In this rule, the properties of the composite depend directly on the volume fractions of each component in the mixture and the respective properties of the fiber and the matrix.

$$E_c = E_f V_f + E_m V_m \quad \text{Isostrain condition}$$

$$E_c = \frac{E_m E_f}{E_m v_f + E_f v_m} \quad \text{Isostress condition}$$

And the stress supported by the composite material is:

$$\sigma_c = \sigma_f V_f + \sigma_m V_m \quad \text{Isostrain condition}$$

$$\sigma_c = \sigma_f = \sigma_m \quad \text{Isostress condition}$$

where σ = strength of the material, V = volume fraction of component, E =modulus of elasticity of the material. The subscripts c is for the finished composite, f is for the fiber, and m is for the matrix. Figure 2a illustrates the result on strength and modulus of the combination of properties in the composite. The effects of increasing the amount of fiber in the composite are illustrated in Figure 2b. At zero reinforcement the composite has the strength of the matrix, while as the content of fibers is increased (volume fraction), it increases in a linear manner that would reach the strength of the fibers at 100% fiber content if this condition were possible.

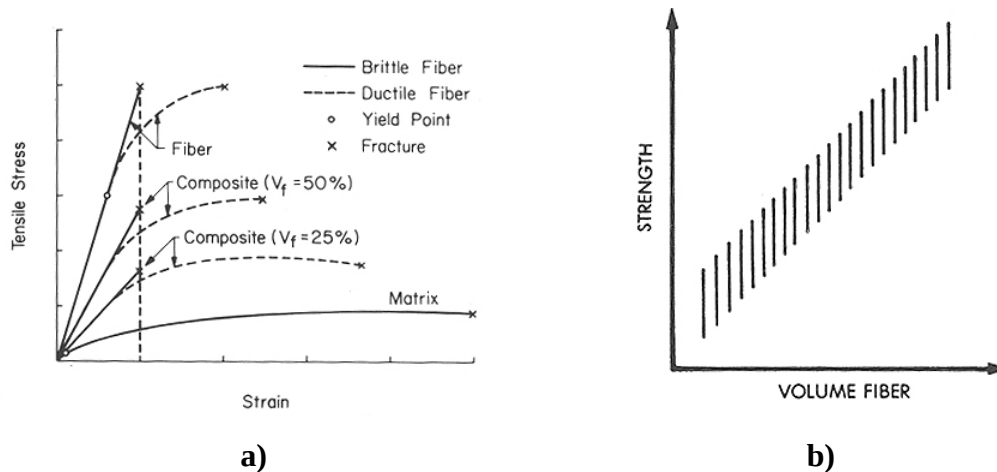


Figure 2.- a) Properties of the composite based on the properties of the component materials; b) the strength of the composite increases with the fiber content.

Fiber orientation also has a great effect on the properties of the composite. Figure 3 illustrates different types of composites based on the fiber orientation: if all the fibers are in one direction, the composite is called unidirectional (a); if the composite has the fibers oriented in all directions, it has randomly oriented fibers (b); if they are in two perpendicular directions, it is called orthogonal or bidirectional (c); and in layers with fibers oriented in different directions (d).

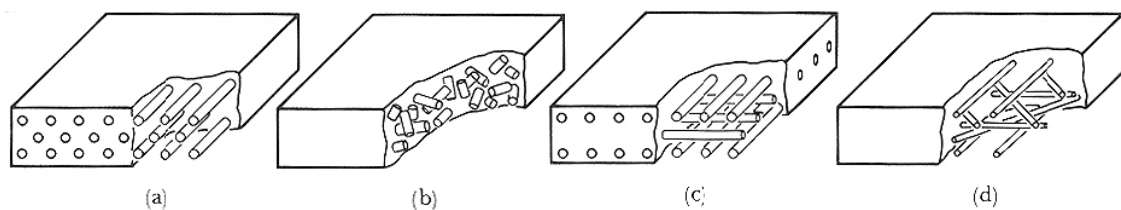


Figure 3.- Different fiber orientations within the composite: a) Unidirectional, b) random, c) bidirectional, and d) in multiple directions for different planes.

The composite response with regards to fiber orientation depends on the direction of loading, as shown in Figure 4. The unidirectional composite will have the greatest strength of all composites with a load aligned to the fibers. In other directions its strengths will decrease quite considerably. For the bi-directional composite, the highest strength will be lower, but it will occur in two directions. In this manner, as the direction of the fibers becomes more evenly distributed throughout the composite, the highest strength will be diminished, but the properties will become more uniform in all loading orientations.

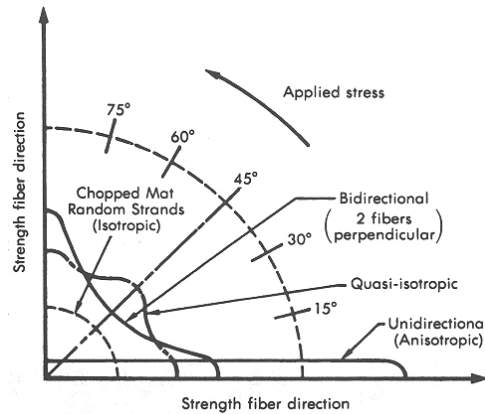


Figure 4.- Dependence of properties based on fiber and loading stress orientations.

The most common type of composites are polymer matrix composites. However, polymer matrix composites have some disadvantages:

- The polymer matrix is relatively weak which affects the final properties of the composite.
- Polymers can only be used in relatively low temperature ranges, else they tend to burn and degrade.
- Polymers often degrade quickly in an outdoor environment.
- Polymers dissipate heat poorly.
- Polymers have high coefficients of thermal expansion.

Because of these and other limitations, interest has risen in fiber reinforced composites, but with metal matrices.

Composite molding methods

In addition to the wide variety of material combinations, there are many choices available for processing composite products. These choices provide for even greater flexibility in optimizing the shape, properties, and production cost of composite components. Matched die molding, contact molding, and more specialized molding methods represent the top-level choices.

Compression molding is the primary choice for most high-volume composite parts. The high-pressure molding process produces highstrength, complex parts in a wide variety of sizes. Matched metal molds are mounted in a hydraulic or mechanical molding press. The material charge is placed in the open mold. The heated mold halves are closed, and pressure is applied. Molding time, depending on part size and thickness, ranges from about one to five minutes. Inserts and attachments can be molded in. (see Figure 5a)

Transfer Compression Molding is characterized by molding operations using a transfer cylinder that is usually built into the tool. The material charge made of resin, reinforcement, and additives is moved to the transfer cylinder and subsequently forced into the closed mold cavity or cavities by the transfer piston. It is best suited for very thick parts, like transformer bobbins, and is very effective for multiple-cavity molds. (see Figure 5b)

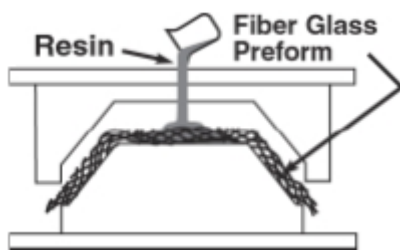
Injection Molding, reinforced thermoset molding compounds can be injection molded in similar equipment commonly used for thermoplastic resins. The principle difference lies in the temperatures maintained in various areas of the system. With thermoplastics, the injection screw and chamber are maintained at a relatively high temperature, and the die is cooled so the molded part sets up. In contrast, for a thermoset, the screw and chamber are cooled so the resin does not cross-link and gel, and the die is heated so it does cross-link and cure. Injection molding offers high-speed production and low direct labor costs combined with excellent mechanical properties. (see Figure 5c)

Hand Lay-Up is the simplest and oldest of the fabrication processes composites. It is used in low-volume production of large components such as boat hulls and associated parts. A pigmented gel coat is first sprayed onto the mold for a high-quality surface. When the gel coat has cured reinforcing material is placed on the mold, and the catalyzed resin is poured, brushed or sprayed on. Manual rolling then removes entrapped air, densifies the composite and thoroughly wets the reinforcement with the resin. Additional layers of material and resin are added for thickness. A catalyst or accelerator initiates curing in the resin system, which hardens the composite without external heat. (see Figure 5d)

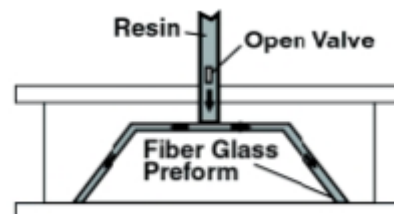
Spray-Up is similar to hand lay-up in simplicity, spray-up offers greater shape complexity and faster production. Chopped fiber reinforcement and catalyzed resin are deposited in the mold from a combination chopper/spray gun. As with lay-up, manual rolling removes entrapped air and wets the fiber reinforcement. Woven roving is often added in specific areas for thickness or greater strength. Pigmented gel coats can be used to produce a smooth, colorful surface. (see Figure 5e)

Vacuum bag processing uses a vacuum to eliminate entrapped air and excess resin from a lay-up form on either a male or female mold. A non-adhering film (usually polyvinyl alcohol or nylon) is placed over the lay-up and sealed at its edges. A vacuum is drawn on the bag formed by the film, and the FRP/Composite is cured, either at room temperature or with moderate heat to speed the process. (see Figure 5f)

Autoclave molding is a further modification of vacuum bag. The process produces denser void-free composites because higher heat and pressure are used in the cure. Autoclaves are essentially heated pressure vessels (usually equipped with vacuum systems) into which bagged lay-ups, on their molds, are cured at pressures of 50 to 100 psi. Autoclaves are normally used to process high-performance components based on epoxy-resin systems for aircraft and aerospace applications.



a)



b)

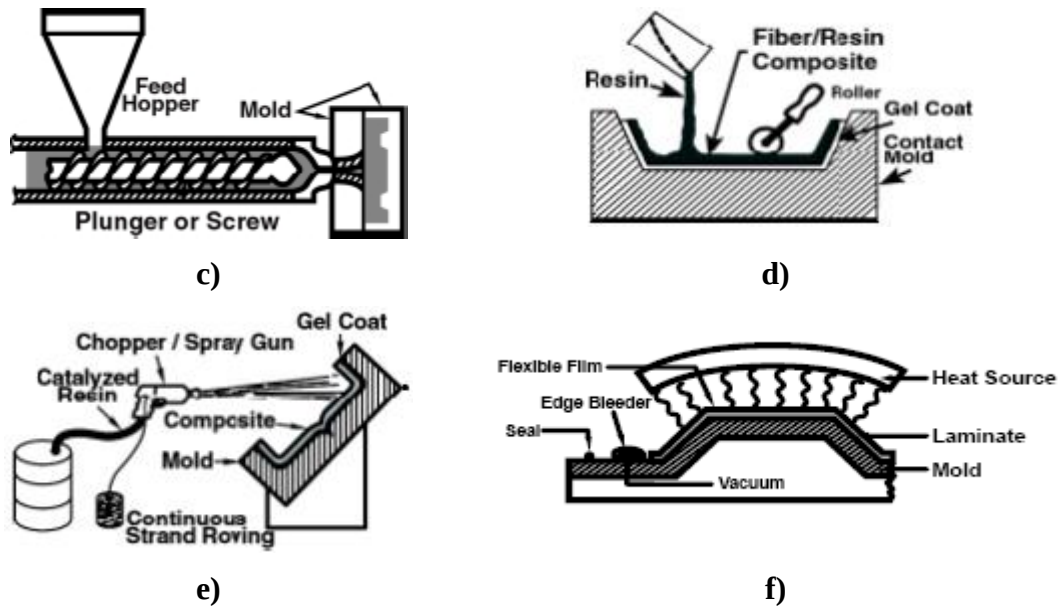


Figure 5.- a) Compression Molding, b) Transfer Compression Molding, c) Injection Molding, d) Hand Lay-Up, e) Spray-Up and f) Vacuum Bag Molding.

OPERATING PROCESS

We are going to build sandwich composite structures using unidirectional tape carbon fiber reinforced epoxy prepreg M21E, usually used for manufacturing primary airframe and aerospace structures i.e. external skin of the plane wing (sandwich and monolithic).

Procedure:

- 1) Extract the laminate from the fridge ($T = -20^{\circ}\text{C}$)
- 2) Separate carefully each laminate of each other
- 3) Cut strips of $150 \times 15 \text{ mm}$ at 0° , 45° and 90° directions (three to each one)
- 4) Take the bottom of the mold and place in position 1, one above other tape cut at direction 0° .
- 5) Repeat step number 4 this time to cut tapes at direction 45° and place in position 2, and at direction 90° and place in position 3.
- 6) Close the mold with top part.
- 7) Place the mold on the bottom plate of the compression machine having attention, it could be hot.
- 8) Slowly bring down the top plate of the compression machine up to the pressure shown in Figure 6.
- 9) Set temperature increases rate until the temperature shown in Figure 6.
- 10) Check that the system is working properly every 5 minutes.
- 11) The total curing time to tape carbon fiber reinforced epoxy prepreg M21E are around 120 minutes.

Note: Work quickly and accurately so as not to affect the final properties of the material.

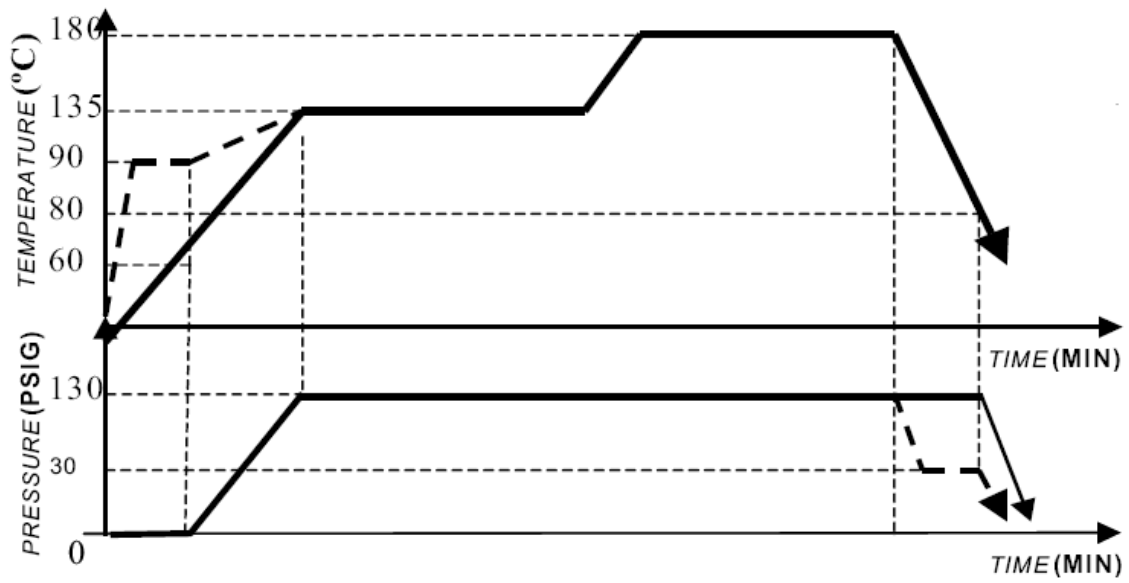


Figure 5.- Autoclave cycle to cured of laminates, manufactured with M21E

Note: 1 Pascal is equal to 0.000145037738007 psig.